GALALA UNIVERSITY
Powered by Arizona State University®

Database Systems

Smart Poultry Farm Management System

CSE221 Database Systems Course Project
Supervised By: Dr. Shaker El-Sappagh
Spring 2025/26

Team:
Omar Khalifa 223101614
Samaa Mohamed 223101629
Mariam EL Zayat 224127559
Basmala Gamal Mohamed 224124339
Ahmed Mohamed Ahmed 224127007

GALALA UNIVERSITY
Powered by
Arizona State University®

Introduction

Smart Poultry Farm Management System Overview

A comprehensive database-driven system designed to manage all aspects of poultry farm operations efficiently.

Traditional farms struggle with tracking inventory, monitoring health, and managing costs accurately.

Automate and organize poultry farm operations through a centralized database system that improves efficiency and decision-making.



GALALA UNIVERSITY
Powered by
Arizona State University®

Project Objectives

System Goals

The Smart Poultry Farm Management System aims to achieve these key objectives for improved farm operations.

- Organize comprehensive farm data in a centralized database for easy access and management.
- Track production cycles efficiently from breed selection through harvest completion.
- Monitor feed consumption and chicken health with daily logging and alerts.
- Manage sales transactions and expenses with accurate financial tracking.
- Support better decision making through reports, dashboards, and AI-powered insights.

GALALA UNIVERSITY
Powered by
Arizona State University®

Business Problem

01

Untracked Behaviour

Manual record-keeping systems lead to frequent data entry errors and processing delays, making it difficult to maintain accurate farm records.

02

Database Anomalies

Tracking individual chickens, monitoring health status, and calculating accurate production costs becomes unreliable without a centralized database system.

03

Security Risks

No proper access control means sensitive farm data is vulnerable, with no way to restrict permissions based on employee roles.

GALALA UNIVERSITY
Powered by
Arizona State University®

Security Features

Why Security Matters

Protecting farm data and ensuring only authorized users can access sensitive information is critical for system integrity.

- **Login Authentication System** - Secure user verification before granting access to the system
- **Password Hashing** - User passwords encrypted using secure hashing algorithms for protection
- **Session Management** - Tracks active user sessions and handles timeouts securely
- **Role-Based Access Control (RBAC)** - Different permission levels for Admin, Manager, and Workers
- **Audit Logs** - Comprehensive tracking of all user actions for accountability and review

GALALA UNIVERSITY
Powered by
Arizona State University®

Smart AI Assistant

01

Ask in Natural Language

Users can ask questions in plain English, such as "How many chickens are in Shed 3?" or "What was last month's feed cost?" No technical knowledge required.

02

Get SQL Results

The AI assistant converts natural language into SQL queries, executes them against the database, and returns accurate results quickly and clearly.

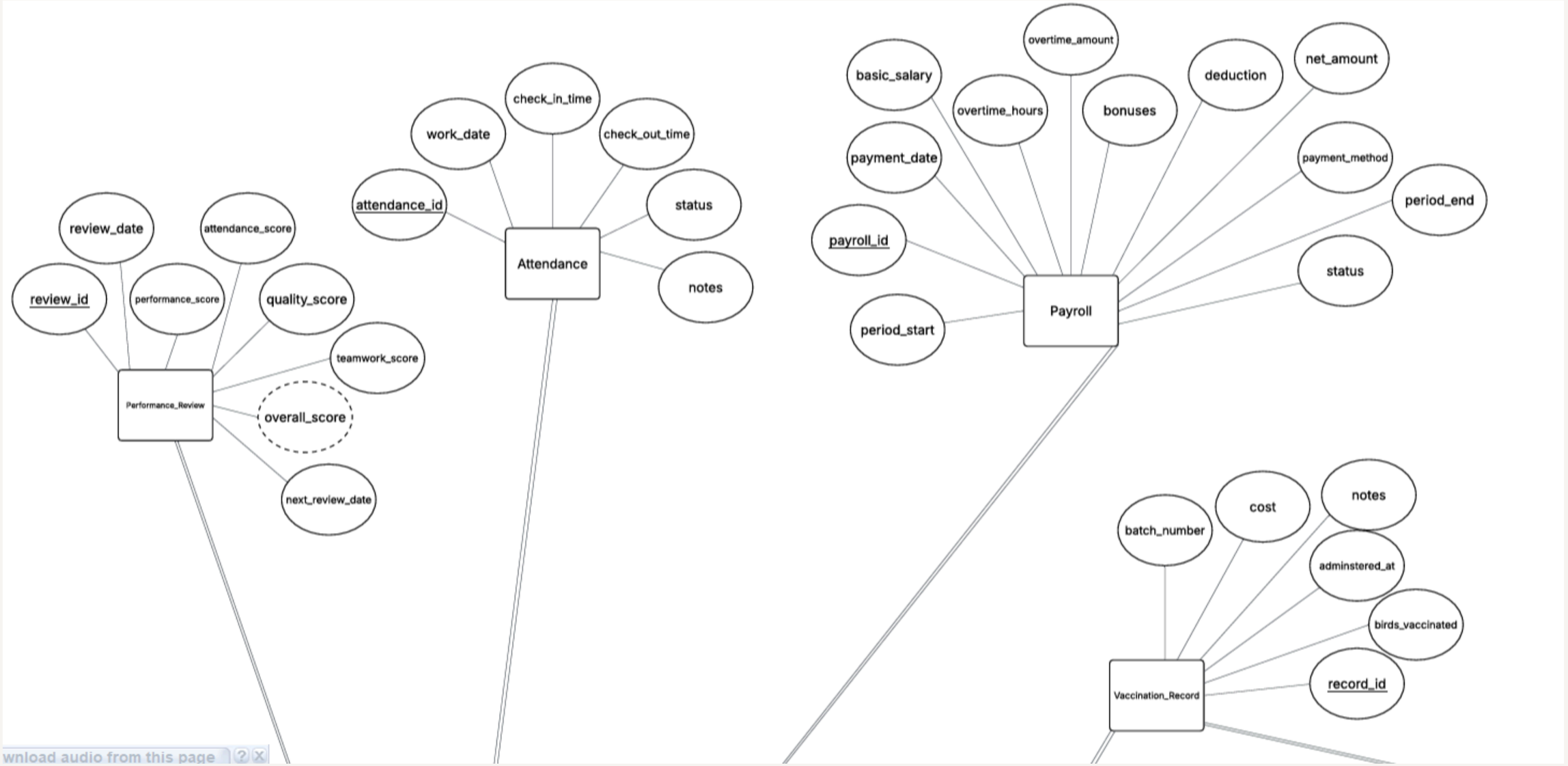
GALALA UNIVERSITY
Powered by
Arizona State University®

Database Design

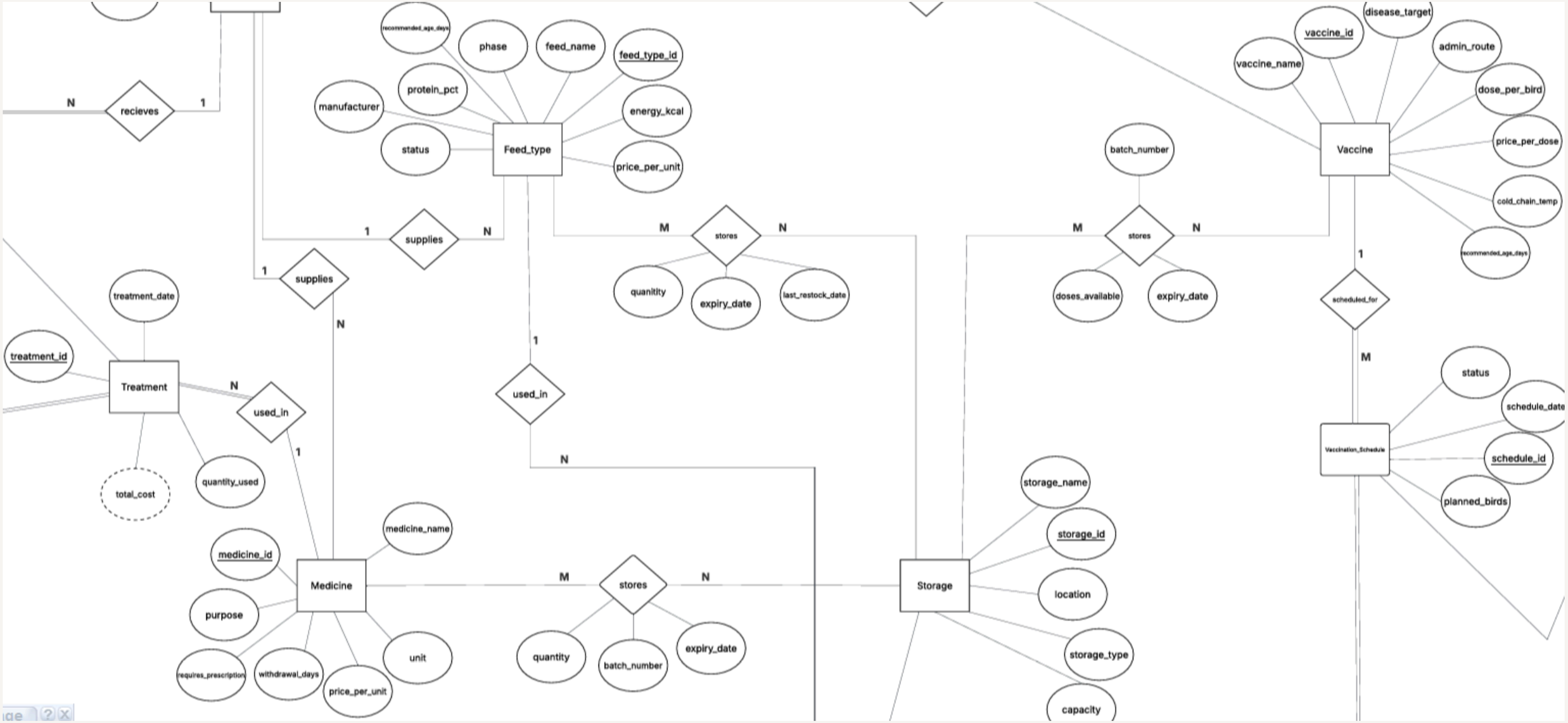
Overview of Database Structure and Normalization

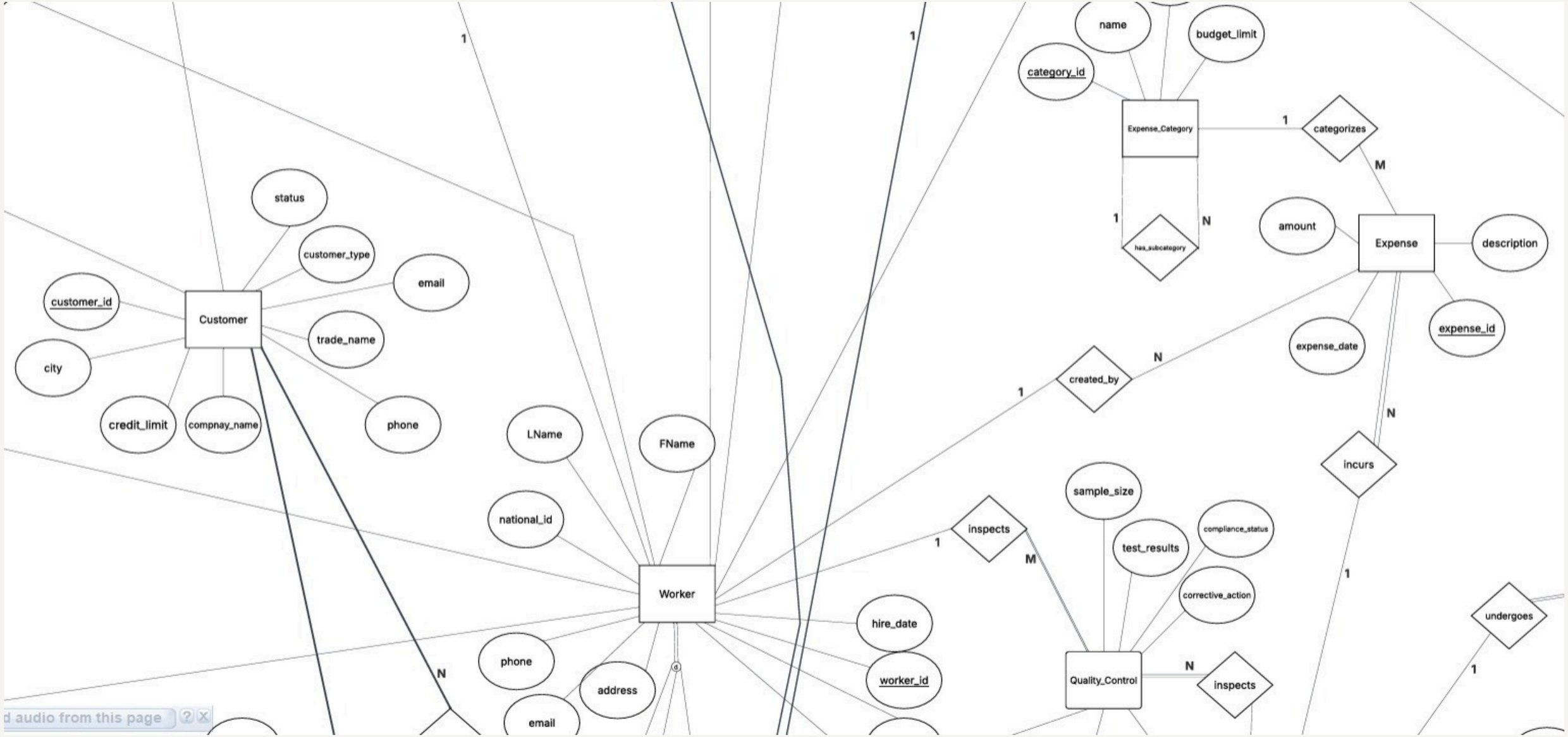
- 35 core entities designed in Third Normal Form (3NF)
- Entity-Relationship Diagram (ERD) maps all relationships
- Primary Keys uniquely identify each record
- Foreign Keys establish links between related tables
- Normalized structure ensures data integrity
- Eliminates redundancy for efficient storage
- Supports complex queries across farm modules

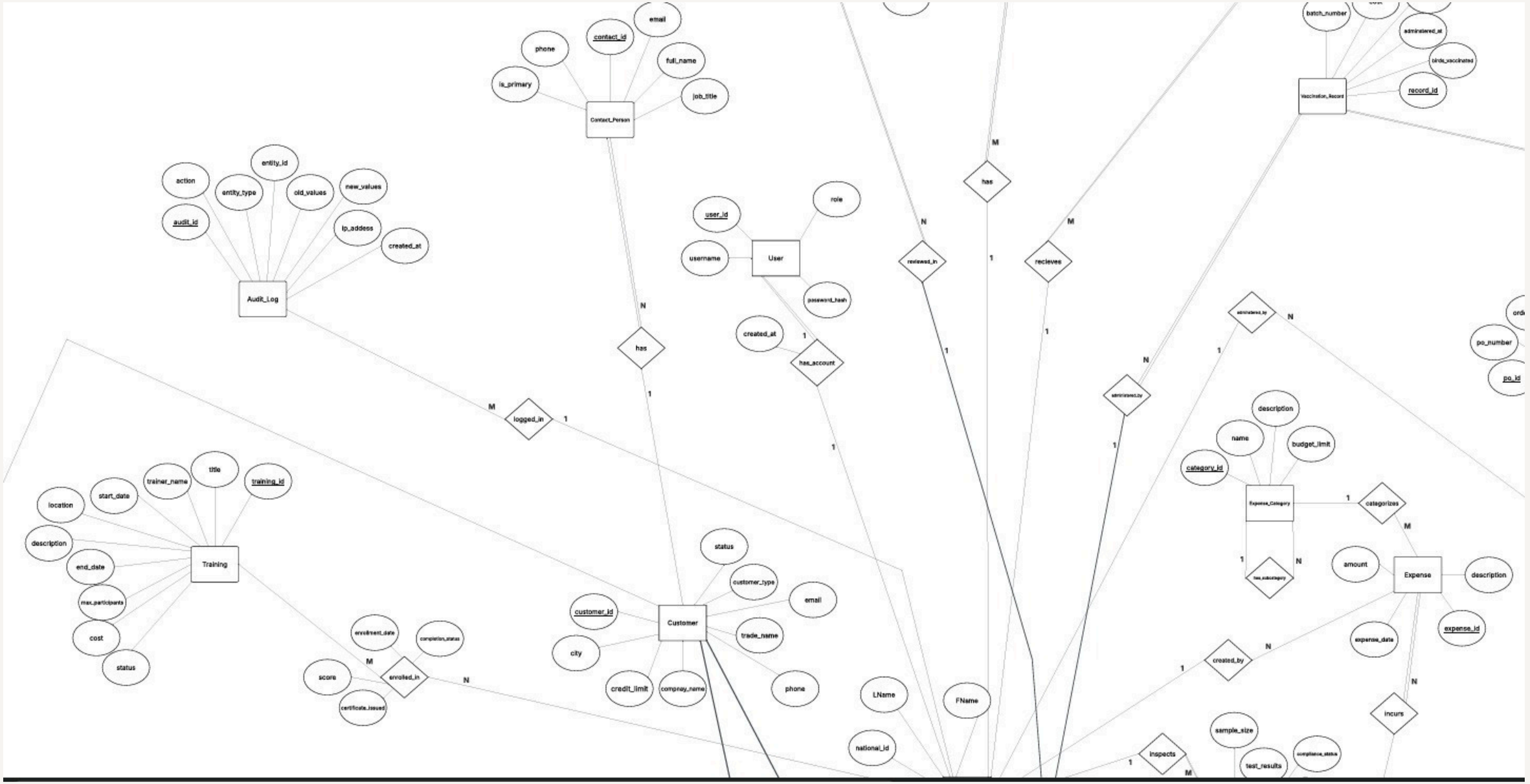
EERD

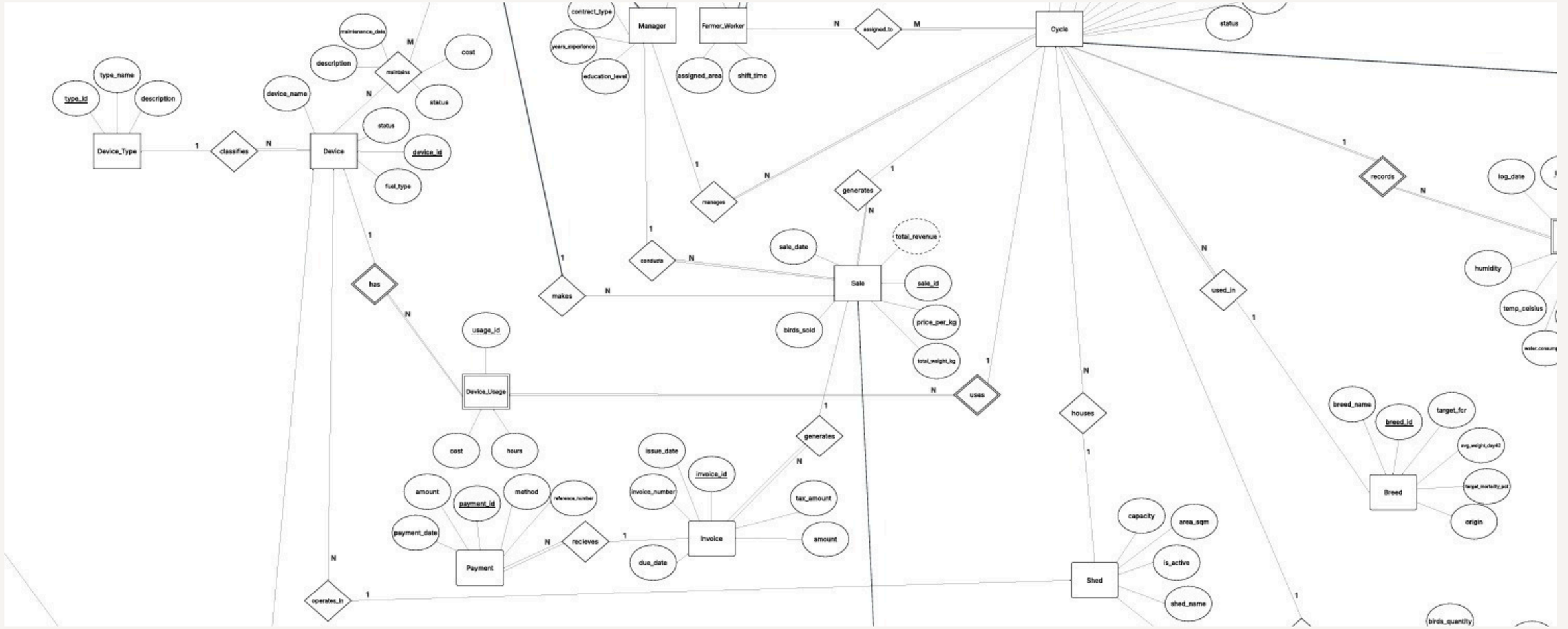


wnload audio from this page


recommended_age_days
phase
feed_name
feed_type_id
protein_pct
manufacturer
energy_kcal
status
Feed_type
price_per_unit
N
recieves
1
1
supplies
N
M
stores
N
quantity
expiry_date
last_restock_date
1
supplies
N
treatment_date
treatment_id
Treatment
N
used_in
1
total_cost
quantity_used
1
used_in
N
medicine_name
medicine_id
Medicine
M
stores
N
purpose
requires_prescription
withdrawal_days
price_per_unit
unit
quantity
batch_number
expiry_date
Storage
storage_name
storage_id
location
storage_type
capacity
batch_number
M
stores
N
doses_available
expiry_date
disease_target
vaccine_id
vaccine_name
admin_route
dose_per_bird
Vaccine
price_per_dose
cold_chain_temp
recommended_age_days
1
scheduled_for
M
status
schedule_date
Vaccination_Schedule
schedule_id
planned_birds


name
budget_limit
category_id
Expense_Category
1
categorizes
M
1
has_subcategory
N
amount
Expense
description
expense_date
expense_id
N
created_by
1
N
incurs
1
sample_size
inspects
1
M
test_results
compliance_status
corrective_action
Quality_Control
N
inspects
undergoes
1
status
customer_type
email
customer_id
Customer
trade_name
city
credit_limit
compnay_name
phone
1
1
LName
FName
national_id
Worker
hire_date
worker_id
phone
address
email
N

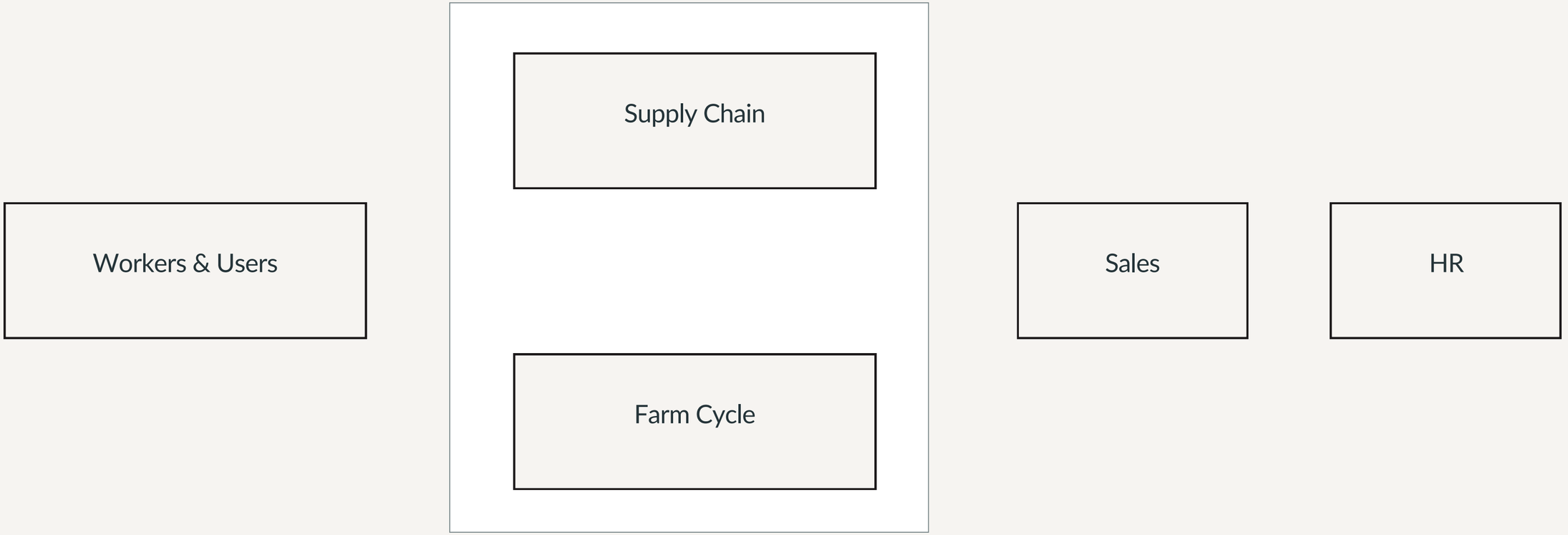

batch_number
administered_at
birds_vaccinated
record_id
Vaccination_Record
email
contact_id
phone
full_name
is_primary
job_title
Contact_Person
has
M
N
1
reviewed_in
recieves
action
entity_id
entity_type
old_values
new_values
audit_id
ip_address
created_at
Audit_Log
user_id
role
username
User
password_hash
created_at
has_account
has
logged_in
administered_by
po_number
po_id
description
name
budget_limit
category_id
Expense_Category
categorizes
has_subcategory
amount
Expense
description
expense_id
expense_date
created_by
incurs
title
trainer_name
training_id
location
start_date
description
end_date
max_participants
cost
status
Training
status
customer_type
email
customer_id
trade_name
city
credit_limit
company_name
phone
Customer
enrollment_date
completion_status
score
certificate_issued
enrolled_in
LName
FName
national_id
inspects
sample_size
test_results
compliance_status


contract_type
Manager
Farmer_Worker
assigned_to
Cycle
status
years_experience
education_level
assigned_area
shift_time
maintenance_date
description
maintains
cost
type_name
type_id
description
device_name
status
status
Device_Type
classifies
Device
device_id
fuel_type
manages
generates
records
log_date
humidity
temp_celsius
has
conducts
sale_date
total_revenue
Sale
sale_id
makes
price_per_kg
birds_sold
total_weight_kg
used_in
usage_id
Device_Usage
uses
houses
breed_name
breed_id
target_fcr
avg_weight_day42
Breed
target_mortality_pct
origin
cost
hours
issue_date
invoice_id
generates
amount
payment_id
method
reference_number
invoice_number
tax_amount
payment_date
recieves
Invoice
amount
due_date
Payment
capacity
area_sqm
is_active
Shed
shed_name
operates_in
birds_quantity

GALALA UNIVERSITY
Powered by
Arizona State University®

Main Database Sections

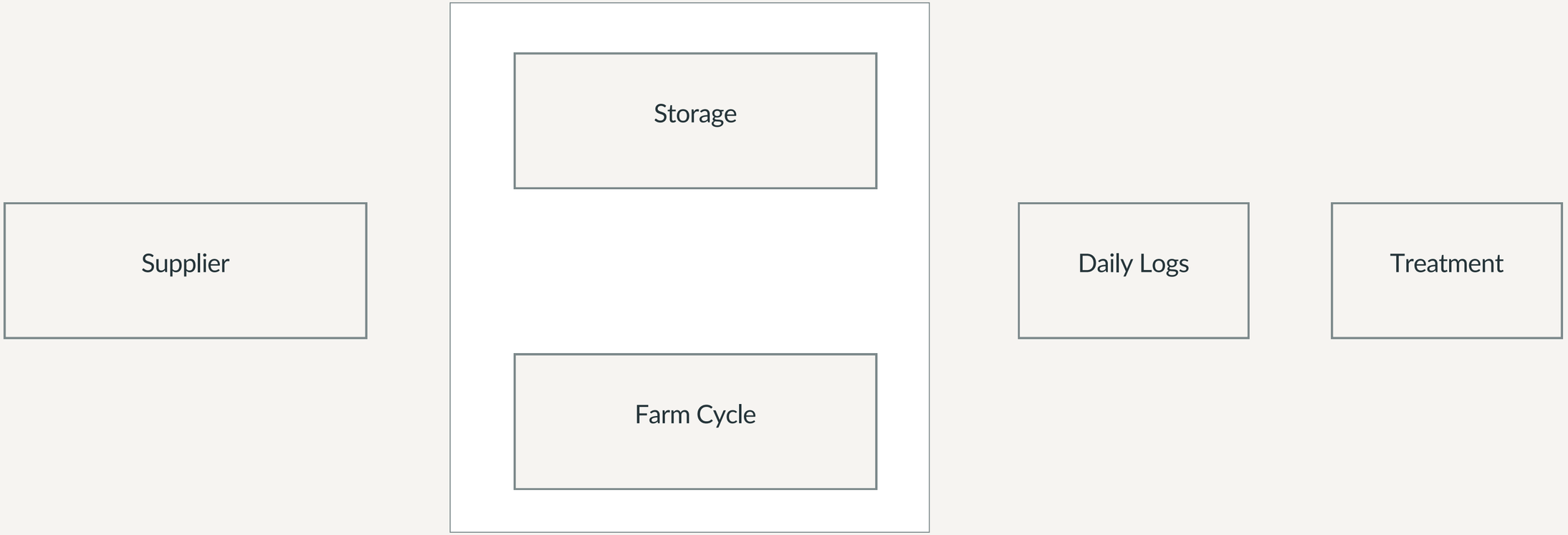




GALALA UNIVERSITY
Powered by
Arizona State University®

Main Database Sections





GALALA UNIVERSITY
Powered by
Arizona State University®

Cycle Management

01 **Breed Selection:** Choose appropriate poultry breeds based on production goals, market demand, and farm capacity requirements.

02 **Shed Assignment:** Allocate birds to designated sheds with proper capacity planning and environmental controls.

03 **Cycle Duration Tracking:** Monitor start and end dates, track growth milestones, and ensure optimal cycle length for maximum productivity.

04 **Daily Logs:** Record daily feed consumption, mortality rates, and vaccination schedules for comprehensive health monitoring.

Triggers & Automation

01



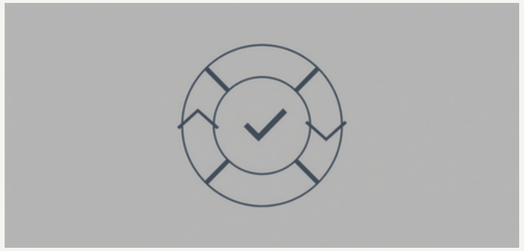
Prevent selling more birds than available stock. Trigger validates inventory before completing any sale transaction.

02



Auto-update inventory after sales. System automatically adjusts stock counts when transactions are recorded.

03



Automatically close production cycles when complete. Trigger detects cycle end conditions and updates status.

GALALA UNIVERSITY
Powered by
Arizona State University®

Frontend / GUI

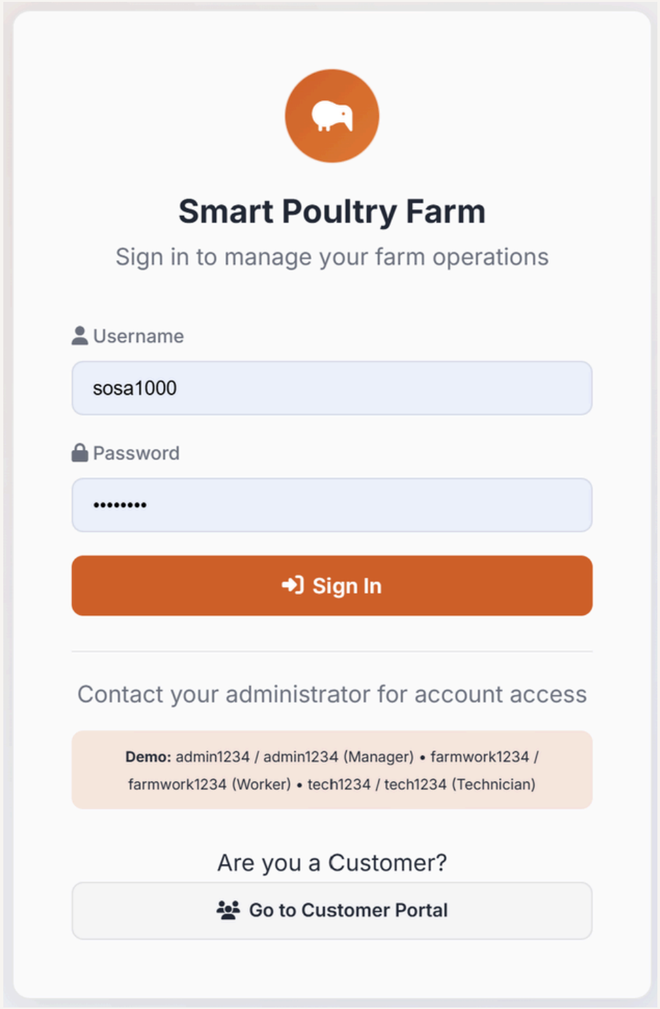
Key system interfaces designed for ease of use:

- Login Page - Secure authentication entry point
- Dashboard Overview - Real-time farm metrics at a glance
- AI Assistant Interface - Natural language query system
- Reports Page - Production and financial summaries
- Daily Logs Form - Input for feed, mortality, health data

Some Pages of GUI

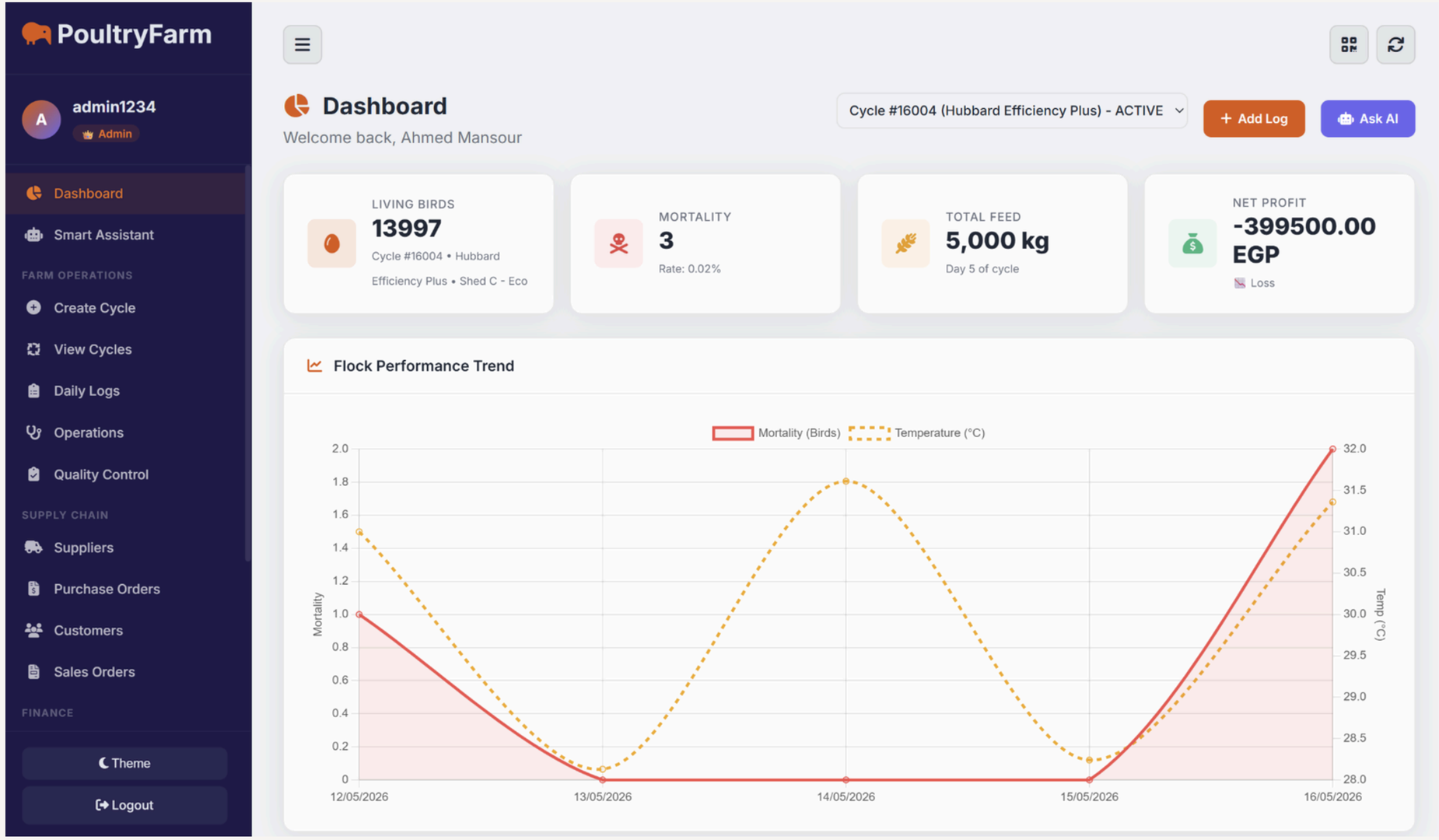
GALALA UNIVERSITY
Powered by
Arizona State University®

Login Page



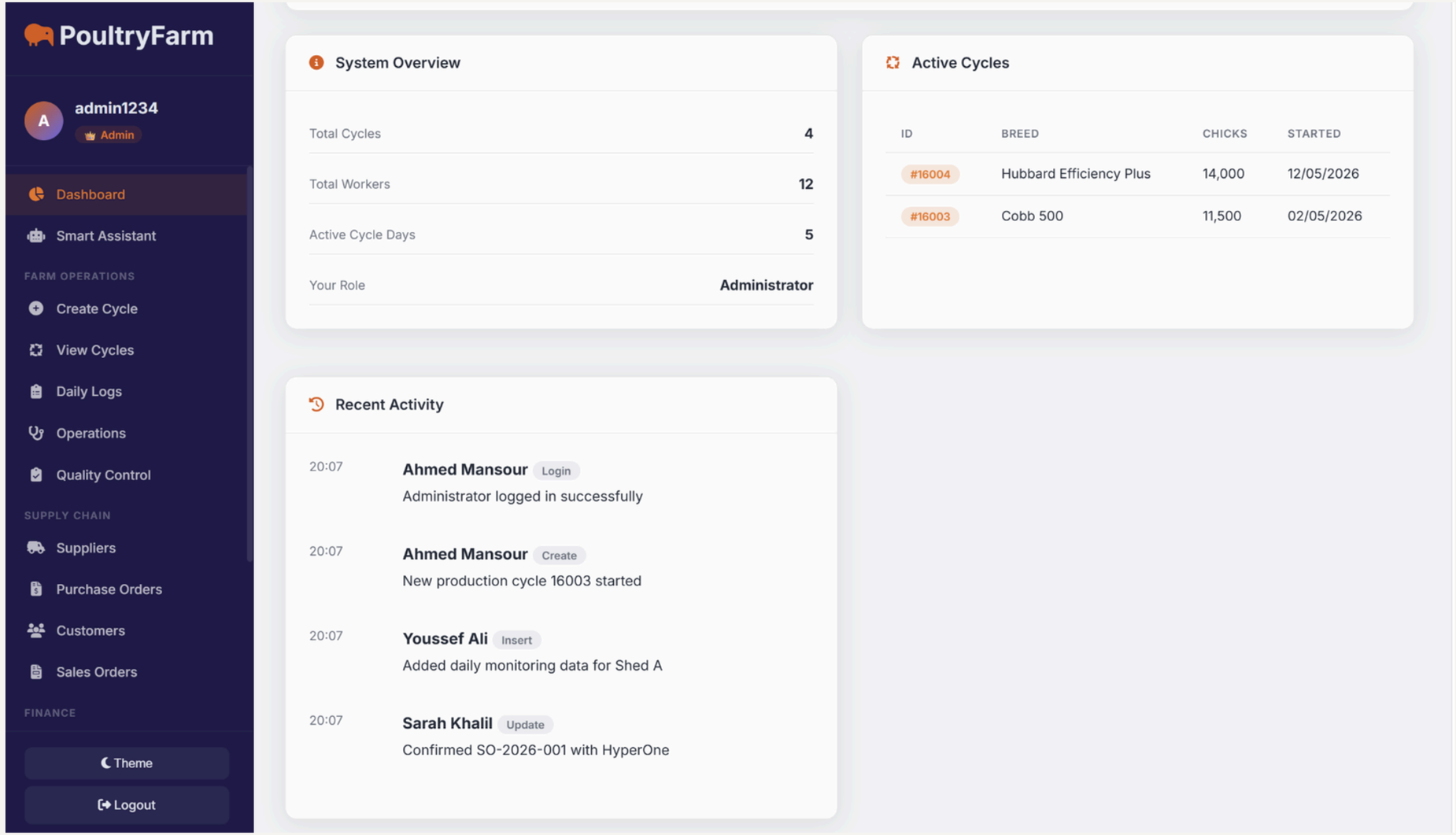
GALALA UNIVERSITY
Powered by
Arizona State University®

Dashboard Overview



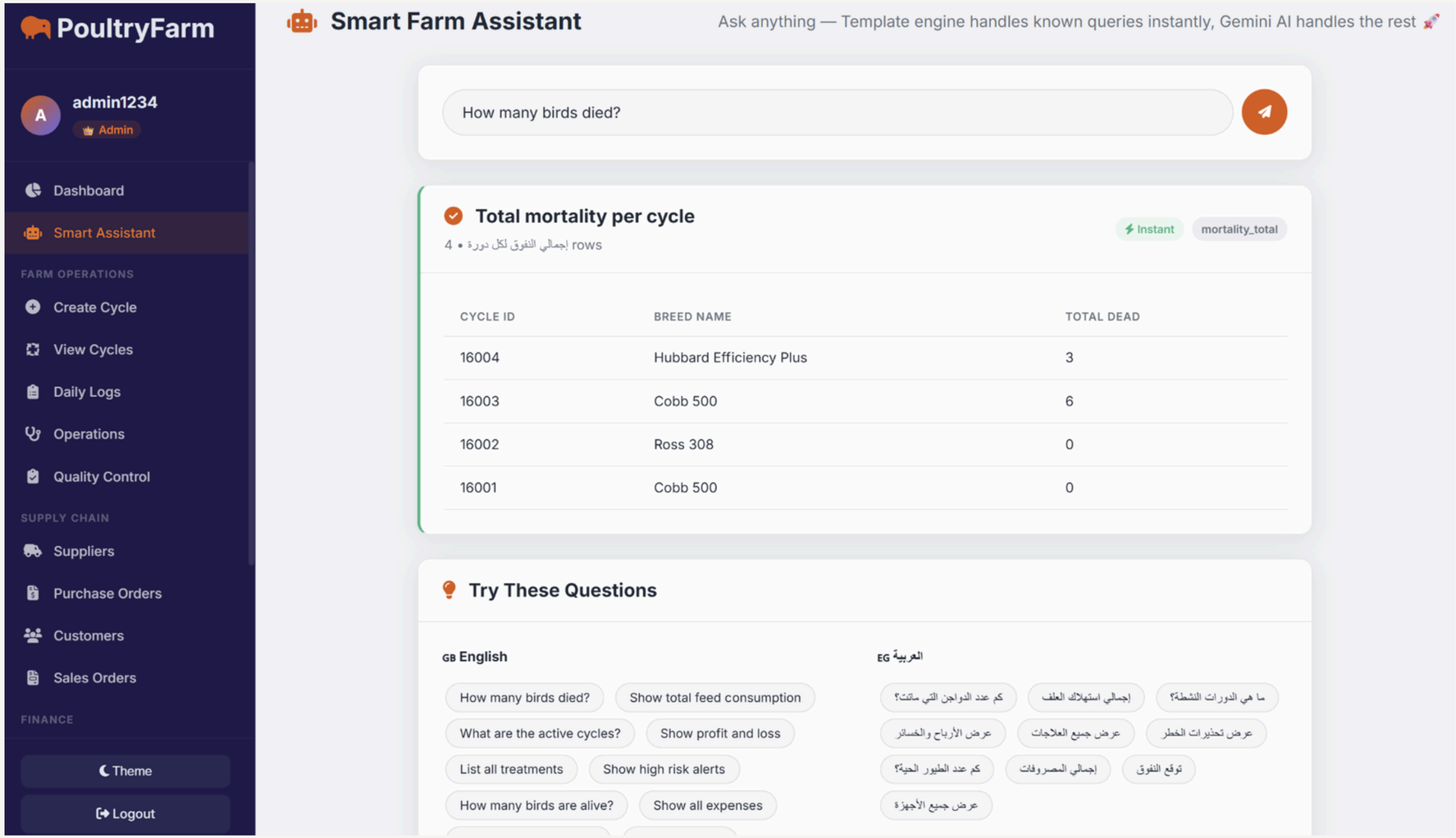
GALALA UNIVERSITY
Powered by
Arizona State University®

Dashboard Overview



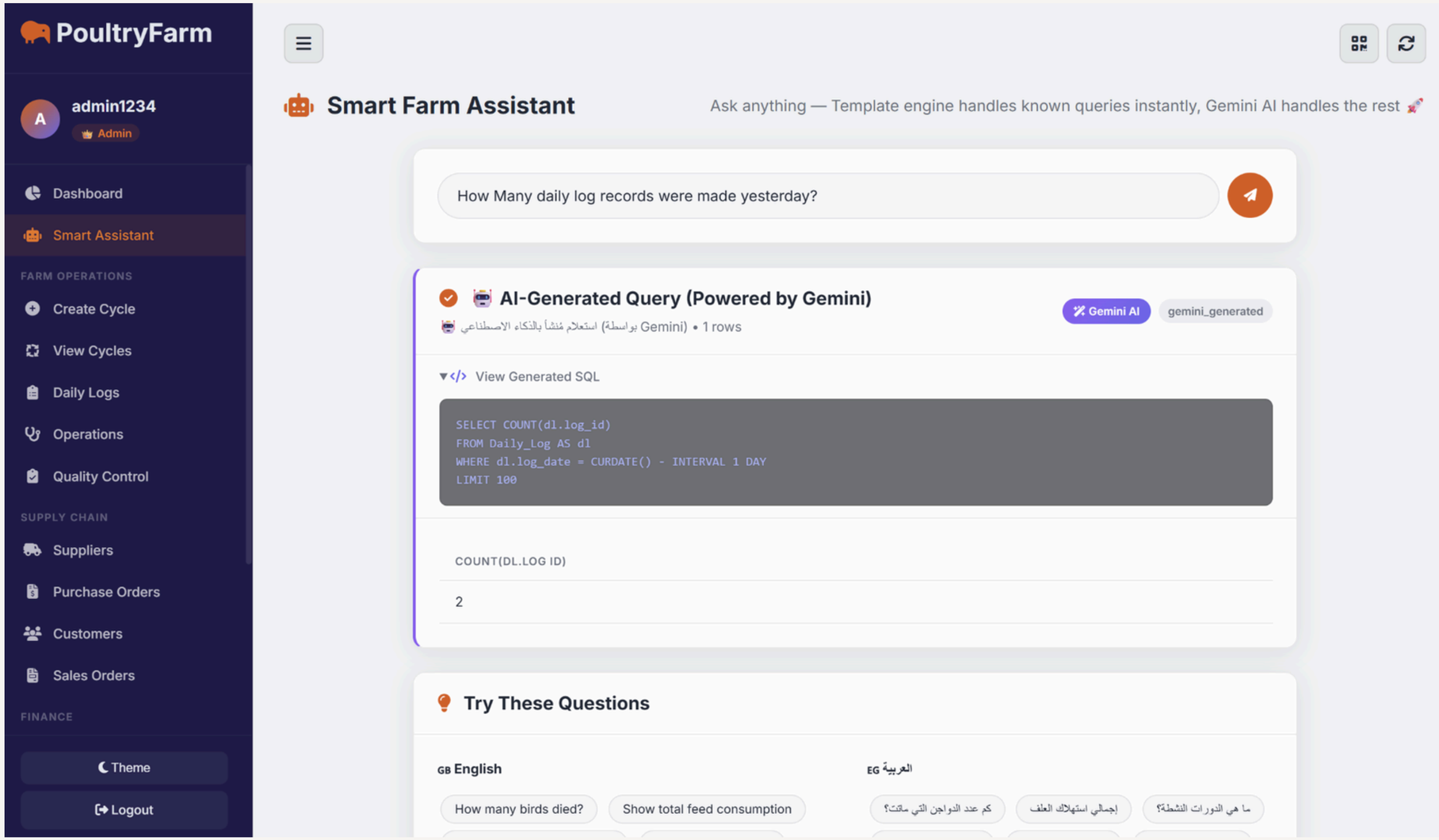
GALALA UNIVERSITY
Powered by
Arizona State University®

AI Assistant Interface - RAG



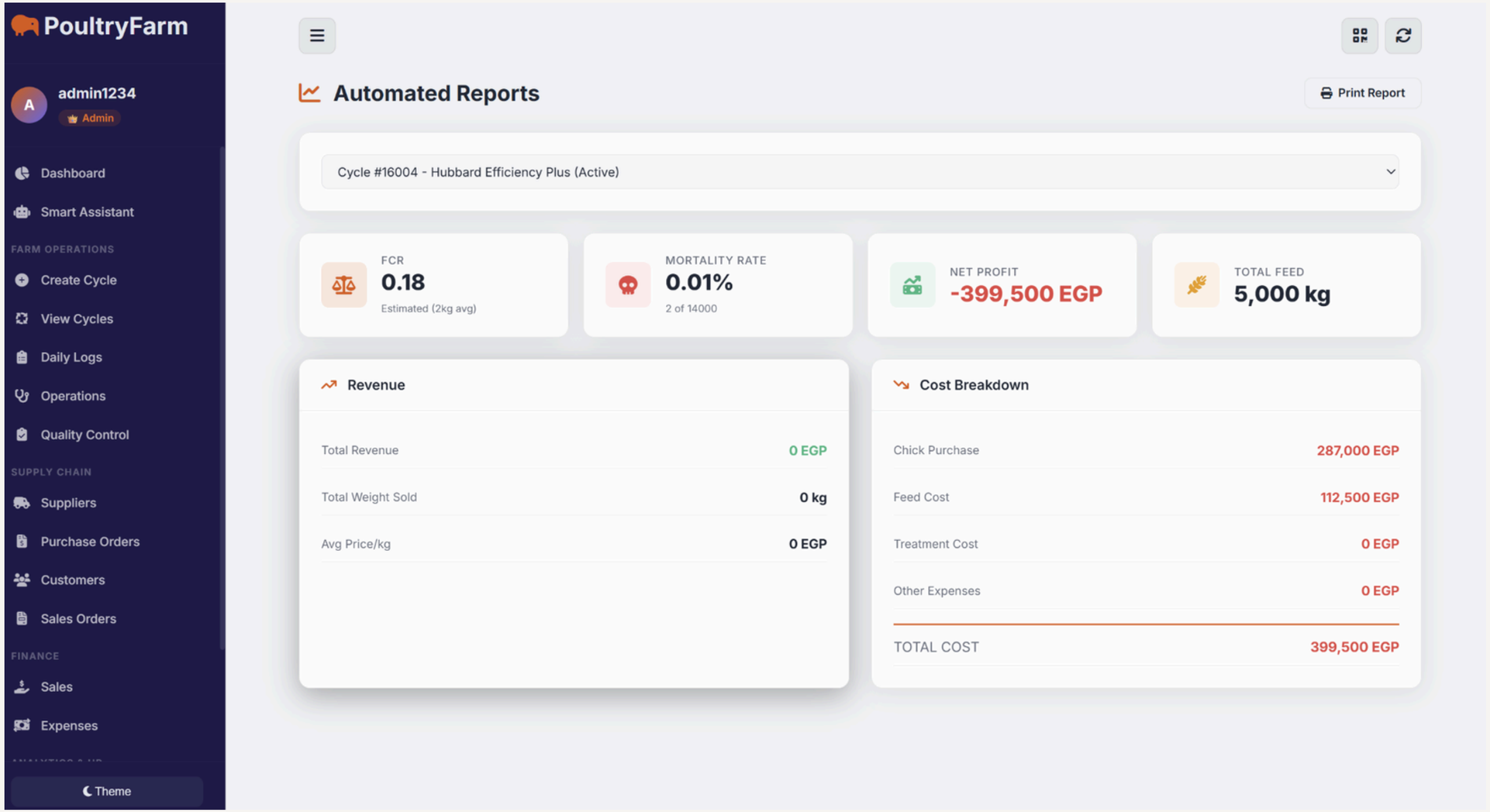
GALALA UNIVERSITY
Powered by
Arizona State University®

AI Assistant Interface - API



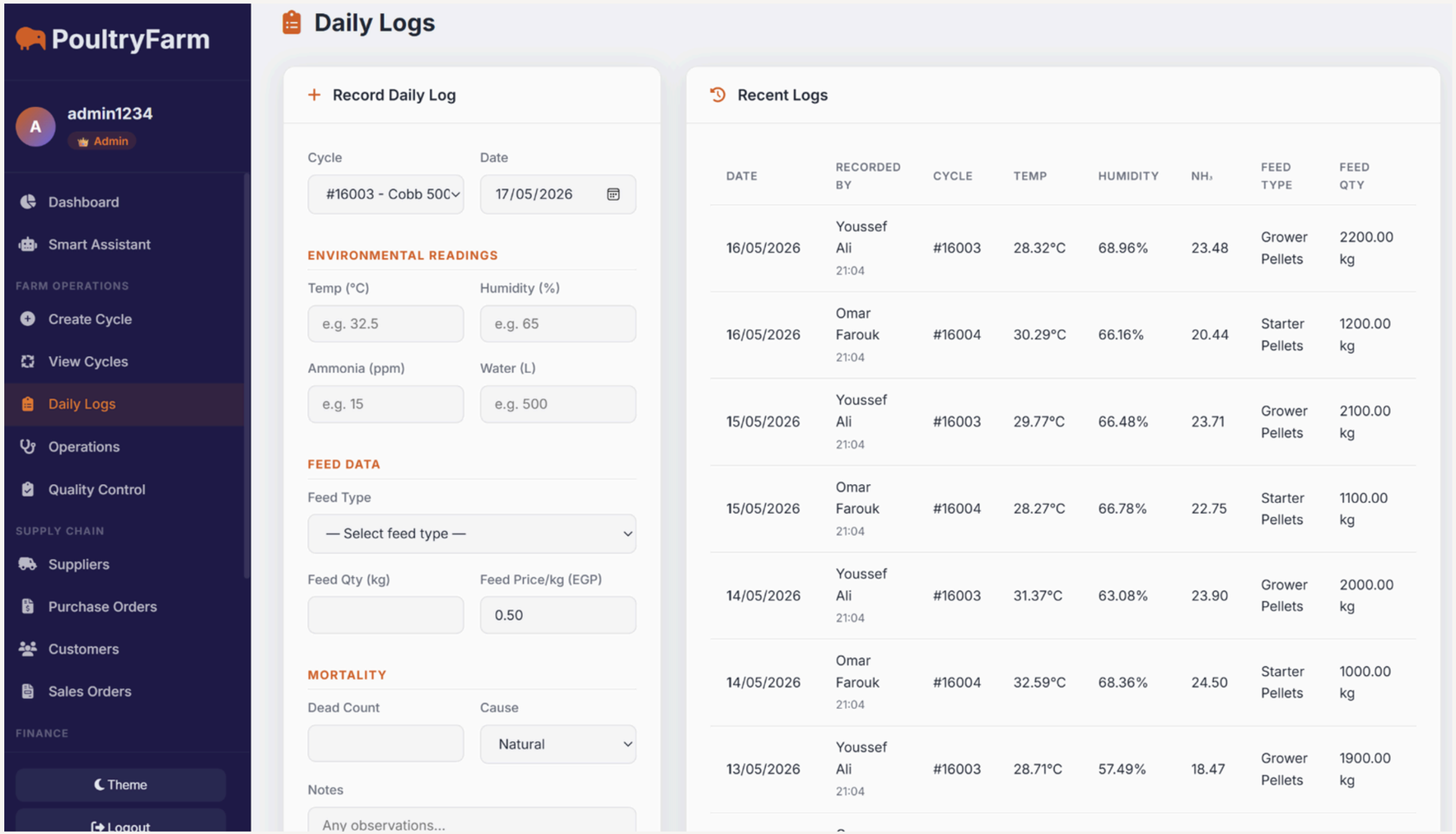
GALALA UNIVERSITY
Powered by
Arizona State University®

Reports Page



GALALA UNIVERSITY
Powered by
Arizona State University®

Daily Logs



GALALA UNIVERSITY
Powered by
Arizona State University®

Challenges
solved

01 Validating inventory data accurately proved difficult due to multiple data entry points and the need to reconcile stock counts across different farm locations and storage units.

02 Handling real-time calculations for feed consumption, mortality rates, and profit margins demanded optimized queries to avoid performance bottlenecks.

03 Managing complex database relationships between 35 entities required careful planning to maintain referential integrity while ensuring efficient query performance.

04 Building effective trigger logic to automate inventory updates, cycle closures, and stock validation without creating conflicts or infinite loops was challenging.

GALALA UNIVERSITY
Powered by
Arizona State University®

Future Improvements

01 IoT sensor integration for real-time data collection on temperature, humidity, feed levels, and environmental conditions.

02 Push notifications and alerts to inform users of critical events like low inventory, health issues, or completed cycles in real-time.

03 Predictive analytics for farm management, using historical data to forecast production outcomes and optimize decisions.

GALALA UNIVERSITY
Powered by
Arizona State University®



Conclusion

The Smart Poultry Farm Management System helps farms operate smarter and more organized. By leveraging database technology and automation, the system streamlines daily operations, reduces manual errors, and supports better decision-making for sustainable farm management.

Smart Poultry Farm Management System

GALALA UNIVERSITY
Powered by Arizona State University®

Thank You

Database Systems - Spring 2026